

Picturing the Floor Plan

Mental-Imagery Vividness Predicts Who Lingers at the Building Directory, Not Signage Clarity

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Introduction

Wayfinding audits treat a lingering visitor as a signage failure. Converting a floor plan into a mental route takes people different amounts of time — is dwell time measuring confusion, or a stable difference in how vividly people picture space?

Research questions

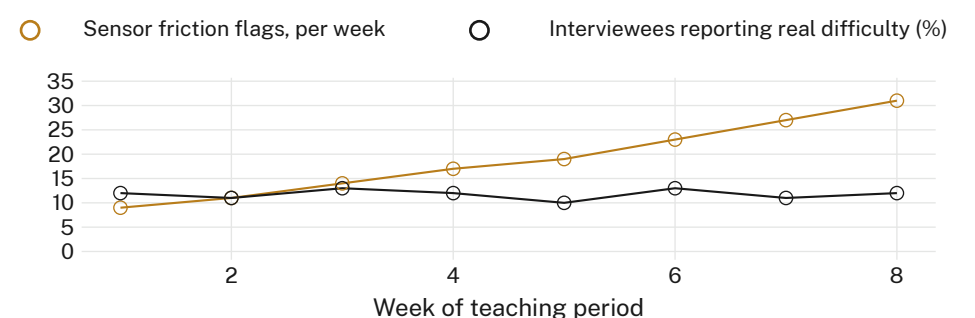
- Does self-reported imagery vividness predict directory-board dwell time?
- Does the sensor's friction flag track genuine wayfinding difficulty?
- Do flagged buildings differ in imagery-vividness mix, not signage quality?

Approach

- n = 166 directory-board pauses intercepted and interviewed · 9 buildings · one teaching period
- VVIQ imagery-vividness self-report plus a 2-minute route-recall task
- Sensor's existing 25-second threshold held unchanged; no recalibration attempted

What we found

Imagery-vividness quartile predicted dwell time more strongly than any building-level signage variable measured. Sensor-flagged buildings reported no more wayfinding difficulty than unflagged ones.



Sensor-issued wayfinding-friction flags rose from 9 to 31 per week across weeks 1–8 of Semester 1, 2026 (n = 166 board pauses); interviewees reporting genuine navigation difficulty held between 10% and 13% throughout.

Discussion

The sensor's threshold is tracking the imagery-vividness mix of who happens to pass, not the board's clarity. A signage redesign aimed at the flag would edit a wall that was never the problem, and the redesign brief now in circulation cites the flag count as its evidence base without once naming the cohort it actually describes. Next: validate the threshold against VVIQ composition before it authorises any redesign spend.

Further reading

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Human ethics approval 2025/118; VVIQ responses de-identified before matching to sensor logs; thresholds pre-registered.

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